

Woei-Chyi Chang
 Postdoctoral Researcher
 Lyles School of Civil and Construction Engineering
 Purdue University,
 West Lafayette, IN 47907
 Email: chang803@purdue.edu
 Website: <https://woeichyi.github.io/>
 Phone: (765) 418-4574

1. EDUCATION & PROFESSIONAL POSITIONS

Postdoctoral Researcher..... Jul 2025–Present
 - Purdue University, United States (Advisor: Dr. Sogand Hasanzadeh)
 Ph.D. in Civil Engineering (*Construction Engineering*) 2021–2025
 - Purdue University, United States (Advisor: Dr. Sogand Hasanzadeh)
 - *Dissertation: Determinants of Successful Worker-Autonomy Teaming in Future Construction Workplaces*
 M.S. in Electrical and Computer Engineering (*Computer Engineering*)..... 2024–2025
 - Purdue University, United States (Advisor: Dr. Lu Su)
 M.S in Civil Engineering (*Construction Management*)..... 2017–2020
 - National Taiwan University (NTU), Taiwan (Advisor: Dr. Po-Han Chen)
 - *Thesis: Application of Mixed Reality and Infrared Inspection to Examination of External Wall Tiles*
 - Study Abroad, Technical University of Darmstadt, Darmstadt, Germany 2019–2020
 B.S. in Civil Engineering 2013–2017
 - National Taiwan University (NTU), Taiwan

2. HONORS & AWARDS

Best Ph.D. Dissertation Award, Purdue University 2025
 The Estus H. and Vashti L. Magoon **Research Excellence Award**, Purdue University 2025
Best Paper Award, World Building Congress (WBC) Conference 2025
 2nd Place Prize in the Poster Competition, ITS Michigan Annual Meeting..... 2025
 Kinnier Dissertation Fellowship, Purdue University 2024
 Crooks Graduate Scholarship, Purdue University 2024
 Invited for Plenary Talk (top 12 papers), ISARC Conference..... 2023
 Invited to Submit Extended Work to Automation in Construction, EC3 Conference 2023
 Lyle School of Civil Engineering Graduate Travel Award, Purdue University 2023
 Academic Choice Runner-up Award in CII Poster Competition, Purdue University..... 2022
 Professional Engineer (P.E.) License in Taiwan 2018

3. RESEARCH EXPERIENCE (*Funding sources: 2 NSF, 1 USDOT, 3 State DOTs, and 3 International agencies*)

Smarter and Safer Work Zone Through Multimodal Intrusion Detection, USDOT..... 2025–Present
 - Role: Leading Postdoctoral Researcher, Purdue University
 Integrated Smart Work Zones, Indiana DOT (INDOT)..... 2023–Present
 - Role: Leading Postdoctoral Researcher/Graduate Research Assistant, Purdue University
 Worker-AI Teaming in Future Construction Industry, FW-HTF National Science Foundation (NSF) 2021–Present
 - Role: Leading Postdoctoral Researcher/Graduate Research Assistant, Purdue University
 Improving Worker Safety by Understanding Risk Compensation as a Latent Precursor of At-risk Decisions,
 Collaborative Research National Science Foundation (NSF)..... 2021–2023

- Role: Graduate Research Assistant, Purdue University
 Implementation and Assessment of Work Zone Intrusion Technologies, INDOT..... 2021–2023
 - Role: Graduate Research Assistant, Purdue University
 Synthesis Study on Best Practices of Cleaning Tools & Paving Equipment: Asphalt Release Agents (ARA) and Asphalt Cleaners (AC), INDOT..... 2021–2022
 - Role: Leading Graduate Research Assistant, Purdue University
 Application of BIM and Feng-Shui to Architectural Design, National Science and Technology Council 2020
 - Role: Leading Graduate Research Assistant, National Taiwan University
 Application of BIM to Support Grading Hosing Performance – Fire Protection, Barrier-free Environment, and Residential Maintenance, Architecture and Building Research Institute..... 2017–2018
 - Role: Leading Graduate Research Assistant, National Taiwan University
 Combining Augmented Reality with UAV for Real-Time Recognition of Rust, National Science and Technology Council 2017–2018
 - Role: Leading Graduate Research Assistant, National Taiwan University

4. PUBLICATIONS

4.1 Peer-Reviewed Published Journal Papers:

- [01] **Chang, W. C.**, & Hasanzadeh, S. (2025). “Mental Workload in Worker-Drone Communication in Future Construction: Considering Coexistence, Cooperation, and Collaboration Interaction Levels”. *Advanced Engineering Informatics*, 65(A), 103110. DOI: <https://doi.org/10.1016/j.aei.2025.103110>
- [02] **Chang, W. C.**, Garcia, N. F. G., Esmaili, B., & Hasanzadeh, S. (2025). “Deep Learning–based Prediction of Human-Robot Trust Dynamics in Future Construction”. *Journal of Computing in Civil Engineering*, 39(4), 04025047. DOI: 10.1061/JCCEE5.CPENG-6302
- [03] **Chang, W. C.**, Esmaili, B., & Hasanzadeh, S. (2025). “The Impacts of Physical and Informational Failures on Worker-Autonomy Trust in Future Construction”. *Journal of Construction Engineering Management*, 151(4), 04025011. DOI: 10.1061/JCEMD4/COENG-15241
- [04] **Chang, W. C.**, Lu, H. K., & Chen, P. H. (2025). “Application of Mixed Reality and Infrared Inspection to Examination of External Wall Tiles”. *Journal of Performance of Constructed Facilities*, 36(6), 04025062. DOI: 10.1061/JPCFEV.CFENG-5179
- [05] **Chang, W. C.**, & Hasanzadeh, S. (2024). “Toward a Framework for Trust Building between Humans and Robots in the Construction Industry: A Systematic Review of Current Research and Future Directions”. *Journal of Computing in Civil Engineering*, 38(3), 03124001. DOI: 10.1061/JCCEE5.CPENG-5656
- [06] Pooladvand, S., **Chang, W. C.**, & Hasanzadeh, S. (2024). “Identifying At-Risk Workers Using fNIRS-Based Mental Load Classification: A Mixed Reality Study”. *Automation in Construction*, 164, 105453. DOI: 10.1016/j.autcon.2024.105453
- [07] Chen, P. H., Huang, T. Y., **Chang, W. C.**, & Lin, Y. H. (2022). “A Smart Scoring Method for The Assessment of Office Lighting Systems”. *Journal of Building Engineering*, 61, 105258. DOI: 10.1016/j.jobbe.2022.105258
- [08] Cheng, M. Y., Yeh, S. H., & **Chang, W. C.** (2020). “Multi-Criteria Decision Making of Contractor Selection in Mass Rapid Transit Station Development Using Bayesian Fuzzy Prospect Model”. *Sustainability*, 12(11), 4606. DOI: 10.3390/su12114606

4.2 Under-review and In-progress Journal Papers:

- [01] **Chang, W. C.**, Karalunas, S. L., Esmaili, B., & Hasanzadeh, S. (2025). “How ADHD Workers Will Read the Future Construction Sites: The impact of time pressure and multitasking on situational awareness, productivity, and stress management”. (*Submitted by December 2025*).
- [02] **Chang, W. C.**, Pooladvand, S., & Hasanzadeh, S. (2025). “Influence of Worker Stance, Orientation, and Risk Zone on Stress During Human–Drone Communication in Construction” (*Submitted by December 2025*).

4.3 Peer-Reviewed Conference Papers:

- [01] **Chang, W. C.**, Weston, N., & Hasanzadeh, S. (2026). “AI-Driven Trajectory Prediction to Support Worker Monitoring in Work Zone Digital Twins”. *Construction Research Congress 2026*. San Antonio, Texas.
- [02] **Chang, W. C.**, Wang, X., Zhu, T., Adetoto, J. D., Su, L., Feng, Y., Esmaili, B., & Hasanzadeh, S. (2026). “Development of an Integrated Smart Work Zone for Proactive Intrusion Detection and Warning”. *Transportation Research Board Annual Meeting*. Washington D.C..
- [03] **Chang, W. C.**, Yu, L. F., Esmaili, B., & Hasanzadeh, S. (2025). “Exploring Worker-Drone Interaction in Mixed Reality: Balancing Distraction and Situational Awareness”. *The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces*. Arles, France.
- [04] **Chang, W. C.**, Pooladvand, S., & Hasanzadeh, S. (2025). “The Impact of Physically Demanding Human-Drone Communication on Worker Mental Stress”. *The CIB World Building Congress*. West Lafayette, Indiana. **[Best Conference Paper Award]**
- [05] **Chang, W. C.**, Wang, X., Zhu, T., Adetoto, J. D., Su, L., Feng, Y., Esmaili, B., & Hasanzadeh, S. (2025). “Digital Twin for Detecting Worker Violations in Road Construction Work Zones”. *International Conference on Computing in Civil Engineering*. New Orleans, Louisiana.
- [06] **Chang, W. C.**, Zhou, W., & Hasanzadeh, S. (2025). “AI-Driven Assistant for Worker Safety in Future Construction: Examining Opportunity and Challenges”. *Conference on Human Factors and Ergonomics Society Annual Meeting*. Chicago, Illinois.
- [07] **Chang, W. C.**, Pooladvand, S., Esmaili, B., & Hasanzadeh, S. (2024). “Worker-Drone Communication and Stress in Construction Workplaces under Varying Safety Conditions”. *International Conference on Computing in Civil Engineering*. Pittsburgh, Pennsylvania.
- [08] **Chang, W. C.**, Garcia, N. F. G., Esmaili, B., & Hasanzadeh, S. (2024). “Partial Personalization for Worker-robot Trust Prediction in the Future Construction Environment”. *International Symposium on Automation and Robotics in Construction (ISARC)*. Lille, France. **[Invited for Plenary Talk]**
- [09] **Chang, W. C.**, Ismael Becerra, J., Karalunas, S. L., Esmaili, B., Yu, L. F., & Hasanzadeh, S. (2024). “Pioneering Research on a Neurodiverse ADHD Workforce in the Future Construction Industry”. *Construction Research Congress 2024*. Des Moines, Iowa.
- [10] **Chang, W. C.**, Ryan, S. M., Hasanzadeh, S., & Esmaili, B. (2023). “Attributing Responsibility for Performance Failure on Worker-Robot Trust in Construction Collaborative Tasks”. *2023 European Conference on Computing in Construction (EC3)*. Crete, Greece. **[Invited for special collection in Automation in Construction]**
- [11] **Chang, W. C.**, Borowiak, A., & Hasanzadeh, S. (2023). “The Importance of Situational Awareness in Future Construction Work: Toward the Effects of Faulty Robots, Trust, and Time Pressure”. *International Conference on Computing in Civil Engineering*. Corvallis, Oregon.
- [12] **Chang, W. C.**, Zhou, W., & Hasanzadeh, S. (2026). “A Human-Centered Digital Twin Framework for Safety-Critical Human-Robot Interaction”. *The ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. Edinburgh, Scotland, United Kingdom. (*under-review*)

4.4 Book Chapter:

- [01] Book Title: “Advances in Human-AI Collaboration”, Chapter: “Human-AI Interaction Fundamentals”, Editors: Vincent G. Duffy, Waldemar Karwowski, and Gavriel Salvendy. ISBN: 978-1-394-26637-1

4.5 Technical Report:

- [01] Hasanzadeh, S., Esmacili, B., Lee, K., Pokharkar, H., Pooladvand, S., & **Chang, W. C.** (2024). “Implementation and Assessment of Highway Intrusion Technologies”. Purdue University & INDOT. Joint Transportation Research Program. DOI: 10.5703/1288284317727
- [02] Hasanzadeh, S., Velay-Lizancos, M., **Chang, W. C.**, Lopez-Arias, M., & Francioso, V. (2022). “Synthesis Study of Best Practices for Cleaning Tools and Paving Equipment: Asphalt Release Agents (ARAs) and Asphalt Cleaners (ACs)”. Purdue University & INDOT. Joint Transportation Research Program. DOI: 10.5703/1288284317381

4.6 Magazine and News Highlight:

- [01] Lyles School of Civil and Construction Engineering (2024). “Human-Robot Teaming in Construction”. *CE Impact Magazine*. (<https://engineering.purdue.edu/CCE/Media/Impact/2024-Fall/humanrobot-teaming>)
- [02] “Understanding Behavior Through VR”. (<https://www.youtube.com/watch?v=poXCbEZ7SaE>)

4.7 Thesis and Dissertations:

- [01] Ph.D. Dissertation (2025); Lyles School of Civil and Construction Engineering; Purdue University.
Title: “*Determinants of Successful Worker-Autonomy Teaming in Future Construction Workplaces*”
- [02] M.S. Thesis (2020); Construction Engineering and Management; National Taiwan University.
Title: “*Application of Mixed Reality and Infrared Inspection to Examination of External Wall Tiles*”

4.8 Other Oral and Poster Presentations:

- [01] **Chang, W. C.**, & Hasanzadeh, S. (2025). “Digital Twin for Smarter and Safer Work Zone: Proactive Intrusion Detection and Warning” Presented in the Poster Competition at the 2025 *Intelligent Transportation Society Michigan Annual Meeting*. [2nd Place Prize]
- [02] **Chang, W. C.**, Yu, L. F., & Hong, S.R. (2024). “Worker-AI Teaming to enable ADHD Workforce Participation in the Construction Industry of the Future” Presented at the 2024 *Learning Multiverse Conference: AI-Enabled Immersive Technologies in Workforce Training & Education*. [Keynote Speaker]
- [03] **Chang, W. C.**, & Hasanzadeh, S. (2024). “Worker-robot Interactions in the Future Construction Industry: Trust-building and Situational Awareness (SA)” Poster Presented at the *Construction Research Congress*.
- [04] **Chang, W. C.**, & Hasanzadeh, S. (2022). “Worker-AI Teaming to enable ADHD Workforce Participation in the Construction Industry of the Future” Presented at the 2022 *Construction Industry Institute Poster Callout*. [Academic Choice Runner-up Award]

5. PROPOSAL WRITING EXPERIENCE

“Enhancing Worker and Driver Safety through Integrating Digital Twins with Multimodal Sensing and Proactive Alerts”, Smart and Connected Communities (S&CC) NSF	2025
- Assisted in developing research ideas and preparing the grant proposal	
“Smarter and Safer Work Zone Through Multimodal Intrusion Detection”, USDOT (Budget: \$1,980,988)	2024
- Assisted in developing research ideas and preparing the grant proposal	
“Integrated Smart Work Zone”, INDOT (Budget: \$654,939)	2023
- Assisted in developing research ideas and preparing the grant proposal	
NSF CAREER Proposal, Human-centered Computing (HCC) NSF	2023

- Assisted in writing and preparing the grant proposal
“Fortifying School Infrastructure via Stakeholder-Centered Digital Twin”, Strengthening American Infrastructure (SAI) NSF (Budget: \$750,000)..... 2022
- Assisted in writing and preparing the grant proposal
“Synthesis Study of Best Practices for Cleaning Tools and Paving Equipment: Asphalt Release Agents (ARAs) and Asphalt Cleaners (ACs)”, INDOT (Budget: \$52,644)..... 2021
- Assisted in writing and preparing the grant proposal

6. TEACHING EXPERIENCE

6.1 Instructor

- Co-Instructor, Purdue University..... Fall 2025
- Course Name: Project Control & Life Cycle Execution of Constructed Facilities (Undergraduate level)

6.2 Teaching Assistant

- Teaching Assistant, Purdue University..... Spring 2023, Fall 2023
- Course Name: Design Build (Undergraduate /Graduate level)
 - Course Name: Project Control & Life Cycle Execution of Constructed Facilities (Undergraduate level)
- Teaching Assistant, National Taiwan University Fall 2018, Spring 2019
- Course Name: Construction Management (Graduate level)
 - Course Name: Build–operate–transfer (BOT) Project Management (Part-time Graduate level)

6.3 Voluntary Teaching

- Volunteer Teaching Assistant for Graduate Smart Construction Course, Purdue University Spring 2023, 2024
- Volunteer Teaching Assistant for Undergraduate Capstone Projects, Purdue University Spring 2023
- Volunteer Teachers for Primary School in Rural Areas, National Taiwan University..... Fall 2015–Spring 2017

6.4 Mentoring

- Graduate Student Research Mentor, Purdue University 2022–Present
- Almass Riaz (Ph.D. student in Civil and Construction Engineering, Purdue University)
 - Muzi Li (Ph.D. student in Civil and Construction Engineering, Purdue University)
 - Chia-Ling Liu (Ph.D. student in Civil and Construction Engineering, Purdue University)
 - Wei Zhou (Ph.D. student in Civil and Construction Engineering, Purdue University)
 - Pretty Elizabeth John (Master student in Civil and Construction Engineering, Purdue University)
- Undergraduate Student Research Mentor, Purdue University 2021–Present
- Nathan Weston (B.S. student in Civil and Construction Engineering, Purdue University)
 - Jocelyn Silva (B.S. student in Civil and Construction Engineering, Purdue University)
 - Nestor Felipe Gonzalez Garcia (Exchange student in Civil Engineering, from University of the Andes, Colombia)
 - Andrew Borowiak (B.S. student in Civil and Construction Engineering, Purdue University)
 - Sophia Marie Ryan (B.S. student in Civil and Construction Engineering, Purdue University)

7. OUTREACH & SERVICE ACTIVITIES

- BUILD, FIX, DRIVE IT: Mini Smart Builders, Imagination Station Children Museum, West Lafayette Fall 2024
- Provided 7-year-old children with opportunities to learn bridge-building, construction robotics, and immerse themselves in a virtual reality experience of the future.

(https://www.linkedin.com/posts/purdueece_earlier-this-month-purdue-university-lyles-activity-7246515563504508929--Hi9?utm_source=share&utm_medium=member_desktop)

- Women in Engineering Program, Purdue University Spring 2023, Spring 2024
 - Demonstrated construction robotics and virtual reality for female high school students, allowing them to tangibly learn the application of cutting-edge technology in construction.
- Grand Opening of Trimble Technology Lab, Purdue University March 1, 2024
 - Showcased the deployment of the MR device, Trimble XR 10, to revolutionize the current construction and industry for the donors and President Chiang. (<https://www.youtube.com/watch?v=-oskQ33f0wI>)
- Homecoming in Civil Engineering, Purdue University September 30, 2023
 - Developed interactive activities for Purdue alumni and industry partners to collaborate with construction robots.
- Dedication of Dudley and Lambertus Halls, Purdue University April 14, 2023
 - Demonstrated the application of virtual reality technology in civil and industrial engineering for the donors. (<https://polytechnic.purdue.edu/newsroom/dudley-lambertus-halls-gateway-complex-dedication-signals-kinship-between-polytechnic>)

8. PROFESSIONAL DEVELOPMENT

8.1 Reviewer in Journals/Conferences/Presentation:

- Journal of Construction Engineering Management 2025
- International Conference on Computing in Civil Engineering (I3CE), New Orleans 2025
- Automation in Construction 2024
- Advanced Engineering Informatics 2024
- IEEE Internet Computing (Digital Twins for Industry 4.0 or Smart Manufacturing) 2024
- International Symposium on Automation and Robotics in Construction (ISARC), Lille, France..... 2024
- International Conference on Computing in Civil Engineering (I3CE), Pittsburgh..... 2024
- Construction Research Congress (CRC 2024), Des Moines..... 2023
- International Conference on Computing in Civil Engineering (I3CE), Corvallis..... 2023
- Summer Undergraduate Research Fellowship (SURF), Purdue University 2023

8.2 Affiliations:

- Student member, ASCE Visualization, Information Modeling, and Simulation Committee (VIMS)..... 2022–Present
- Student member, American Society of Civil Engineers (ASCE) 2022–Present